FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

JEE (Advanced)-2019

PAPER – 1

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. AC

Sol. $F = \frac{16kQqx}{d^3} = \frac{4Qqx}{\pi\varepsilon_o d^3}$

$a = -\frac{F}{m} = -\frac{4Qqx}{\pi\varepsilon_p md^3}$; $T = 2\pi\sqrt{\frac{\pi\varepsilon_o md^3}{4Qq}}$

2. BC

Sol. As $\lambda_0 = \frac{hc}{eV}$ and as V increases. λ_0 decreases but characteristic wavelengths do not change so interval between $\lambda_{k\alpha}$ & λ_0 increases and the same for the interval between $\lambda_{k\beta}$ & λ_0

3. ABC

Sol. e = a

And $\rho = \frac{m}{v}$

4. ABD

Sol. Bernoulli's theorem for an orifice at depth 'x' in liquid '3d'.

$P_o + \left(dg\frac{H}{2} + 3dg.x\right) = P_o + \frac{1}{2}\times 3d \times v^z$ …(1)

$\frac{H}{2} - x = \frac{1}{2}gt^2$ …(2)

$R = Vt$ …(3)

Solve for R and apply maxima/minima.

5. ABCD

Sol. Due to rotation no potential difference will be across the end.

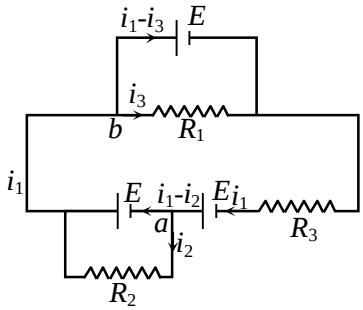
6. B

Sol. Consider three more identical surfaces, such that a square shaped cylinder is formed with point charge at the centre. Flux will be equally divided between these surfaces.

7. B

8. A

Sol. (for Q. 7 & Q. 8)



On solving by KVL we get, $i_1 = \frac{E}{R}$, $i_2 = -\frac{E}{R}$

$i_3 = \frac{E}{R}$

$\Rightarrow\ i_1 - i_2 = \frac{2E}{R}$ and $i_1 - i_3 = 0$

So reading of ammeter A_1 is E/R = 4/3

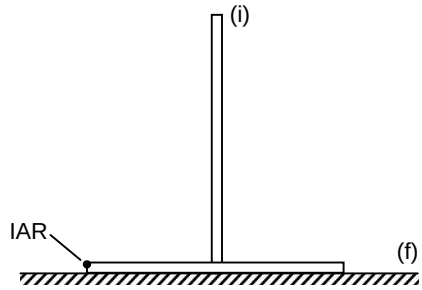
A_2 is $2E/R$ = 8/3, A_3 is zero and ΔV_{ab} = 4V

9. B

Sol. $K_f = mg\frac{\ell}{2}$

and $\frac{1}{2}\ m\frac{\ell^2}{3}\omega^2 = mg\frac{\ell}{2}$

$\Rightarrow\ \omega = \sqrt{\frac{3g}{\ell}}$



10. B

Sol. $K_f = mg\frac{\ell}{2}$

SECTION – D

11. 00125.00

Sol. Energy required = $\frac{1}{2}mv^2 = \frac{1}{2}\times 10\times 5\times 5$ = 125 J.

12. 00009.00

Sol. $a = \mu g$

$6^2 = 0^2 + 2 \times \mu g S$

13. 00015.00

Sol. $V_B \cos\theta_B = V_A \cos\theta_A \Rightarrow V_B = 15\,m/s$

14. 00000.60

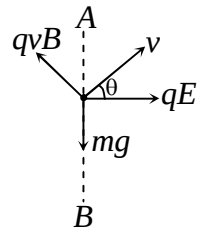
Sol. $\frac{2}{3}MR_1^2 = \frac{2}{5}MR_2^2$

15. 00002.50

Sol. EPE per unit volume = $\frac{1}{2} \times \text{stress} \times \text{strain}$

16. 00019.60

Sol. When crosses *AB*



$qvB_0 \cos\theta = mg$

$qvB_0 \sin\theta = qE$

$\tan\theta = \frac{qE}{mg} = 1$

$\theta = \frac{\pi}{4}$

along horizontal $v\cos\theta = \frac{qE}{m}t_0$

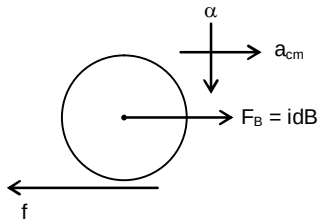
$u - gt_0 = v\sin\theta$

$u = \left(g + \frac{qE}{m}\right)t_0 = 2gt_0$

17. 00002.00

Sol. $f = \frac{F}{3} = \frac{idB}{3}$

$\Rightarrow \frac{48 \times 0.5 \times 0.25}{3} = 2.00$



18. 00000.39

Sol. Applying Snell's law between the points *O* and *P*, we have

$2 \times \sin 60^0 = \left(\sin 90^0\right) \times \frac{2}{\left(1+H^2\right)}, \quad 2 \times \frac{\sqrt{3}}{2} = 1 \times \frac{2}{\left(1+H^2\right)}$

$\left(1+H^2\right) = \frac{2}{\sqrt{3}}, \quad H = \sqrt{\left(\frac{2}{\sqrt{3}} - 1\right)}$

Chemistry

PART – II

SECTION – A

19. ABC
Sol. PCl_5 has trigonal pyramidal structure.

20. ACD
Sol. The ratio of [salt] to [acid] or [base] should be 1 : 10 or 10 : 1 for buffer solution.

21. BCD
Sol. $CH_3CH = CH_2$ will form alcohol upon hydrolysis.

22. BCD
Sol. Silicones contain carbon, silicone and oxygen. Silanes and selenes contain silicone and hydrogen.

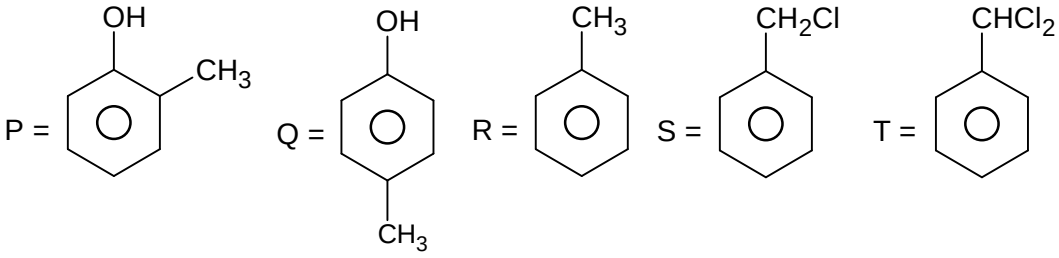
23. B
Sol. It contains no complex ions.

24. ABC
Sol. The precipitate is $Fe_4[Fe(CN)_6]_3$.

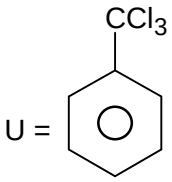
25. C
Sol. $D = \frac{\text{mol.mass of } SO_3}{2} = \frac{80}{2} = 40$

26. C
Sol. $\alpha = \frac{40 - 30}{(1.5 - 1)30} = \frac{10}{15} = 0.66$

27. D
Sol.



28. B
Sol.



SECTION – D

29. 00348.16

Sol. $I.E_4 \text{ of } Be = \left(\frac{Z^2}{n^2} \times 13.6 \times 1.6 \times 10^{-19}\right) J$

$= 16 \times 13.6 \times 1.6 \times 10^{-19} = 348.16 \times 10^{-19}$

$= Y \times 10^{-19}$ J

$\therefore$ Y = 348.16

30. 33088.40

Sol. $\Delta H = nC_P\Delta T = 5 \times 4R \times (1727.21 - 900)$

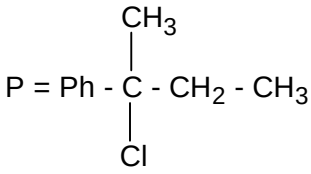
$= 5 \times 4 \times 2 \times 827.21 = 33088.40$ Cal

31. 00124.71

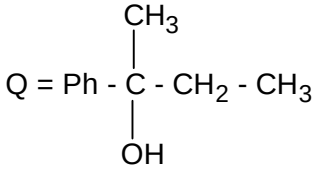
Sol. $K.E = \frac{3}{2}nRT = \frac{3}{2} \times 1 \times 8.314 \times 10^{-3} \times 10000 = 124.71\,kJ$

32. 00168.50

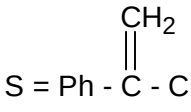
Sol.



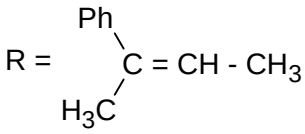
P = Ph - C(Cl)(CH3) - CH2 - CH3



Q = Ph - C(OH)(CH3) - CH2 - CH3



S = Ph - C(=CH2) - CH2CH3

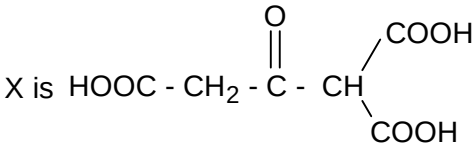


R = Ph, H3C, C = CH - CH3

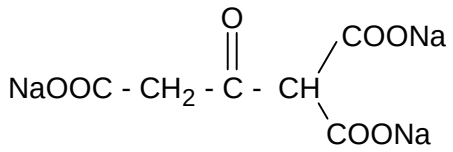
T = $PhCOCH_3$, U = CH_3CHO

33. 00235.50

Sol.



X is HOOC - CH2 - C(=O) - CH(COOH)(COOH) $\xrightarrow{\Delta} CH_3COCH_3 + 3CO_2$



NaOOC - CH2 - C(=O) - CH(COONa)(COONa) + 3 CH_3COCl → Anhydride

E.g. $RCOONa + CH_3COCl \xrightarrow{Py} RCOOCOCH_3 + HCl$

34. 00158.50

Sol. P = $CrCl_3$

Q = $Cr(OH)_3$

R = $[Cr(OH)_4]^-$

S = Na_2CrO_4

T = $BaCrO_4$

35. 00444.50

Sol. $Cu^{2+} + 2e^- \longrightarrow Cu$

2F deposits one mole

14 F will deposit 7 mole

Mass of Cu = $7 \times 63.5 = 444.5$ g

36. 00684.60

Sol. Moles of HCl = $\frac{M \times V}{1000} = \frac{5 \times 3.26 \times 1000}{1000} = 16.3$

Moles of $MgCO_3$ = $\frac{16.3}{2} = 8.15$

Mass of $MgCO_3 = 8.15 \times 84 = 684.6$ g

Mathematics

PART – III

SECTION – A

37. ABCD

Sol. $(gof)(x) = (2x - \sin x)^{1/3}$

Range of (gof) (x) is R.

38. BC

Sol. $\lim\limits_{x\to\infty} \sqrt{x}\left(\sqrt{x+1} - \sqrt{x}\right)$

$= \lim\limits_{x\to\infty} \dfrac{\sqrt{x}}{\sqrt{x+1}+\sqrt{x}} = \lim\limits_{x\to\infty} \dfrac{1}{\sqrt{1+\dfrac{1}{x}}+1} = \dfrac{1}{2}$

Option (A): $\lim\limits_{x\to0} \dfrac{\ln(1+x)-x}{x^2} = \lim\limits_{x\to0} \dfrac{\left(x - \dfrac{x^2}{2} + \dfrac{x^3}{3} - \ldots\right) - x}{x^2} = -\dfrac{1}{2}$

Option (B): $\lim\limits_{x\to0} \dfrac{1-\cos x}{x^2} = \lim\limits_{x\to0} \dfrac{2\sin^2\dfrac{x}{2}}{4\times\left(\dfrac{x}{2}\right)^2} = \dfrac{1}{2}$

Option (C): $\lim\limits_{x\to0} \dfrac{\sqrt{1+x}-1}{x} = \lim\limits_{x\to0} \dfrac{(1+x)^{1/2} - (1)^{1/2}}{(1+x)-1} = \dfrac{1}{2}$

Option (D): $\lim\limits_{x\to0} \dfrac{\sqrt{x}}{\sqrt{x+\sqrt{x^2+2x}}} = \lim\limits_{x\to0} \dfrac{1}{\sqrt{1+\sqrt{1+\dfrac{2}{x}}}} = \dfrac{1}{\sqrt{2}}$

39. AD

Sol. $f(x) = \int_0^x \left(1 + |1-t|\right) dt,\ x > 2$

$= \int_0^1 (2-t)\,dt + \int_1^x t\,dt = \dfrac{1}{2}x^2 + 1$

We have, $f(x) = \begin{cases} \dfrac{1}{2}x^2 + 1, & x > 2 \\ 5x + 1, & x \le 2 \end{cases}$

$\therefore \lim\limits_{x\to2^+} f(x) = 3$ and $\lim\limits_{x\to2^-} f(x) = 11$

Hence, f(x) is not continuous at x = 2.

$Rf'(2) = \lim\limits_{h\to0} \dfrac{f(2+h) - f(2)}{h}$

$= \lim\limits_{h\to0} \dfrac{\dfrac{1}{2}(2+h)^2 + 1 - 11}{h}$

$$= \lim_{h\to 0} \frac{(2+h)^2 - 20}{2h} \to \pm\infty$$

40. AC

Sol. $$f(x) = \begin{vmatrix} n & n+1 & n+2 \\ {}^{n}P_{n} & {}^{n+1}P_{n+1} & {}^{n+2}P_{n+2} \\ {}^{n}C_{n} & {}^{n+1}C_{n+1} & {}^{n+2}C_{n+2} \end{vmatrix} = \begin{vmatrix} n & n+1 & n+2 \\ n! & (n+1)! & (n+2)! \\ 1 & 1 & 1 \end{vmatrix}$$

Applying $C_2 \to C_2 - C_1$ and $C_3 \to C_3 - C_1$, we get

$$f(x) = \begin{vmatrix} n & 1 & 2 \\ n! & n.n! & (n^2+3n+1).n! \\ 1 & 0 & 0 \end{vmatrix} = n!(n^2+n+1)$$

41. AB

Sol. $9\cos^{12} x + \cos^2 2x + 1 - 6\cos^6 x \cos 2x - 6\cos^6 x + 2\cos 2x = 0$

$\Rightarrow \quad (3\cos^6 x - 1 - \cos 2x)^2 = 0$

$\Rightarrow \quad \cos^2 x(3\cos^4 x - 2) = 0$

$\Rightarrow \quad \cos x = 0$

$x = n\pi + \frac{\pi}{2}, n \in I$

or $\cos^4 x = \frac{2}{3}$

$\cos x = \pm\sqrt[4]{\frac{2}{3}}$

$x = m\pi \pm \cos^{-1}\sqrt[4]{\frac{2}{3}}, m \in I$

42. ABD

Sol. Let any point on the line through P be $x = \alpha + r\cos\theta,\ y = 2 + r\sin\theta$.

Put this in $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and $xy = 0$.

$PA \times PD = PB \times PC$

$\Rightarrow\ 2\alpha\sin 2\theta + 5\cos 2\theta = 13$

$\Rightarrow\ 13 \le \sqrt{4\alpha^2 + 25}$

43. C

44. D

Sol. (for Q. 43 & Q. 44)

Let $b_k = \frac{1}{a_k}$.

$b_1, b_2, b_3, \ldots$ are in A.P. with $b_1 = \frac{1}{5},\ b_{20} = \frac{1}{25}$

If d is the common difference of this A.P. then $19d = = \frac{1}{25} - \frac{1}{5} = -\frac{4}{25}$

$d = -\frac{4}{475}$

$b_n = b_1 + (n-1)d = \frac{1}{5} - \frac{4(n-1)}{475} = \frac{99-4n}{475}$

$\Rightarrow\ a_n = \frac{475}{99-4n}$

Note that a_n is maximum if $99 - 4n > 0$ and $99 - 4n$ is least

$\Rightarrow\ n < \frac{99}{4} \Rightarrow n \le 24$

Thus, a_n is maximum for $n = 24$.

Also, $a_n < 0$ if $99 - 4n < 0$

$\Rightarrow\ n > \frac{99}{4} \Rightarrow n \ge 25$

Thus, least value of n for which $a_n < 0$ is 25.

45. C

46. B

Sol. (for Q 45 & Q. 46)

The normals at A and B are perpendicular. Hence, AB is a focal chord. Use

(1) The directrix is perpendicular to the line joining the mid-point of a focal chord and the point of intersection of tangents or normals at the ends of the focal chord.

(2) The focus is the foot of the perpendicular from the point of intersection of tangents to the focal chord.

SECTION – D

47. 00001.62

Sol. $2 < x^2 < 3\ \Rightarrow\ \{x^2\} = x^2 - [x^2] = x^2 - 2$

Also, $\sqrt{2} < x < \sqrt{3}$

or $\frac{1}{\sqrt{3}} < \frac{1}{x} < \frac{1}{\sqrt{2}}$

$\therefore\ \left\{\frac{1}{x}\right\} = \frac{1}{x}$

Hence, $x^2 - 2 = \frac{1}{x}$

or $x^3 - 2x - 1 = 0$

or $(x+1)(x^2 - x - 1) = 0$

$\therefore\ x = \frac{1+\sqrt{5}}{2} \quad (\because x > 0)$

48. 00008.00

Sol. For $x > 0$, $\sin x < x$

$\Rightarrow \quad \dfrac{\sin x}{x} < 1$

For $x < 0$, $\sin x > x$

$\Rightarrow \quad \dfrac{\sin x}{x} < 1$

Hence, $\lim\limits_{x\to 0}\left(\left[\dfrac{100x}{\sin x}\right]+\left[\dfrac{99\sin x}{x}\right]\right) = 100 + 98 = 198$

49. 00005.24

Sol. Given, $f(x)+f(y)=f\left(\dfrac{x+y}{1-xy}\right)$

Putting x = 0, y = 0, we get f (0) = 0 ……..(i)

and putting y = –x, we get

$f(x)+f(-x)=f(0)=0$

$\therefore \quad f(x)=-f(-x)$ ……..(ii)

Now, $f'(x)=\lim\limits_{h\to 0}\dfrac{f(x+h)-f(x)}{h}$

$=\lim\limits_{h\to 0}\dfrac{f(x+h)+f(-x)}{h}$ [from Eq. (ii)]

$=\lim\limits_{h\to 0}\dfrac{f\left(\dfrac{x+h-x}{1+(x+h)x}\right)}{h} \quad =\lim\limits_{h\to 0}\dfrac{f\left(\dfrac{h}{1+x^2+xh}\right)}{\left(\dfrac{h}{1+x^2+xh}\right).(1+x^2+xh)}$

$\Rightarrow \quad f'(x)=\dfrac{2}{1+x^2}$ $\left(\because \lim\limits_{x\to 0}\dfrac{f(x)}{x}=2\right)$

$\therefore f'(-2)=\dfrac{2}{1+4}=\dfrac{2}{5}$

and $f(x)=2\tan^{-1}x+c$

$f(0)=0+c=0$

$\therefore \quad c = 0$

$\therefore \quad f(x)=2\tan^{-1}x$

$\Rightarrow \quad f\left(\sqrt{3}\right)=2\tan^{-1}\sqrt{3}=\dfrac{2\pi}{3}$

50. 00001.00

Sol. $\sin\left(\dfrac{x}{2}\right)-1=x^2-2\pi x+\pi^2$

$\Rightarrow \sin\left(\dfrac{x}{2}\right)-1=(x-\pi)^2$

LHS ≤ 0 and RHS ≥ 0.

Then solution is obtained if LHS = RHS = 0 and $x=\pi$ satisfies both.

51. 00007.00

Sol. The given expression can be interpreted as the square of the distance between the points $(\tan A, \cot A)$ and $(\cos B, \sin B)$.

The minimum value of this distance is the minimum distance between the curves $xy = 1$ and $x^2 + y^2 = 1$.

52. 00002.33

Sol. $a(x^2 - x + 1) - (x^2 + x + 1) \geq 0$

$\Rightarrow a \geq \dfrac{x^2 + x + 1}{x^2 - x + 1}$ or $a \geq \dfrac{(x^2 - x + 1) + 2x}{x^2 - x + 1}$

$\Rightarrow a \geq 1 + \dfrac{2x}{x^2 - x + 1}$ or $a \geq 1 + \dfrac{2}{\left(x + \dfrac{1}{x}\right) - 1}$

Now, for $x \geq 2$, the minimum value of $x + \dfrac{1}{x}$ is $\dfrac{5}{2}$.

$\Rightarrow a \geq \dfrac{7}{3}$

53. 00007.00

Sol. $x_1 + x_2 + x_3 = 12$

$t_1 + t_2 + t_3 = 9$	Total solutions	$(t_i = x_i - 1)$
$y_1 + t_2 + t_3 = 3$	Extra solutions	$(y_1 = t_1 - 6)$

54. 00006.12

Sol. Total number of ways in which papers of 4 students can be checked by seven teachers $= 7^4$

Now, number of ways of choosing 2 teachers out of 7 is ${}^7C_2 = 21$

Number of ways in which 4 papers can be checked by exactly two teachers $= 2^4 - 2 = 14$

$\therefore$ Favourable ways $= (21)(14)$

$\therefore$ Required probability $= \dfrac{(21)(14)}{7^4} = \dfrac{6}{49}$.